

Team 1

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Engineering Service Learning

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Element C:

Requirements are listed in priority order, with the most important in the beginning.

1. The lock must securely keep the restroom stall door closed for a minimum of five minutes without failing. Research on the average restroom usage time and interview responses showed that a few minutes of secure closure is reasonable for normal restroom use. The adult user handling the lock agreed that a five-minute minimum was acceptable and confirmed that no longer duration minimum was required. This requirement is of high priority because if the bathroom lock cannot keep the door shut for at least five minutes, the child may not have finished using the restroom, and could then cause difficulties.
2. There are multiple primary stakeholders including TSA agents throughout airport security. The TSA agents confirm that maintaining one-handed low force adult use was essential for safety and portability by complying with the carry-on regulations. It is also important that the lock does not contain any very sharp materials that could stop the lock by TSA and render it useless. This criteria is of very high priority because if the lock does not pass TSA, the client is unable to use the lock.
3. The lock must be attachable and detachable from the outside of the stall in 5 seconds or less. The stakeholder sources indicated that managing a child requires a quick and easy process, without many distractions. Feedback from the parent operating the lock

confirmed that attachment times greater than 5 seconds were impractical. The client reiterated that this requirement was very important to the functionality of the lock.

4. The lock must fit within a standard front pants pocket with maximum dimension of 9.1 inches by 6.4 inches by 1.5 inches and weigh no more than 3 pounds. The stakeholder interview revealed that the lock must be carried during travel and daily use, making portability, limited size, and a weight cap necessary. This requirement was reviewed by the adult user, who confirmed that pocket-sized portability was necessary and approved the size and weight limits. The client showed that he would greatly prefer if the lock design would fit into his front pants pocket. While this does not take away from the functionality of the lock, it would make the client much more satisfied with the end result.
5. The lock must withstand at least 30 uses. Interview data showed that the lock would be reused during multiple trips and restroom visits, requiring durability and resistance to pulling or shaking. From the stakeholder perspectives such as the guardian confirmed that great durability is required. This requirement is of high priority because if the lock is not durable enough, the child cannot use the bathroom for the ending of the trip.
6. The lock must be operable with one hand without losing any effectiveness. This is a requirement that was established during our interview. This requirement is from the user of the lock so this appeals to the primary stakeholder. This requirement is very important to the client, but it theoretically is not mandatory for the functionality of the lock.
7. The lock must prevent a restroom stall door from opening while allowing an adult to operate it from the outside using less than 5 pounds of force. This design requirement is based on stakeholder interview data and ADA research, which shows that adults often

need to be able to operate restroom hardware with limited force. The client has also specifically requested this design feature. This is of lower priority, because it doesn't contribute to the lock's overall ability to lock the stall, but not meeting this requirement would make the client unsatisfied with the product.

8. The lock must be made out of materials that are available in the engineering classroom or materials that are readily available. This requirement is of low priority, but it contributes to the low cost of the final design.